

Robust Zero-Shot Coreset Selection via Density-Aware Subspace Sampling

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Abstract

Coreset selection aims to identify a compact subset of training data that preserves the predictive performance of models trained on the full dataset. Recent zero-shot approaches, such as Iterative Subspace Sampling (ZCore), eliminate the need for labels and repeated optimization by leveraging pretrained feature representations. However, diversity-driven selection strategies are inherently sensitive to noise, often prioritizing outliers due to their geometric uniqueness.

In this work, we propose a robust extension to ZCore by incorporating a density-aware filtering mechanism prior to subset selection. Our method estimates local sample density using k -nearest neighbor statistics and removes isolated points that are likely to be noisy. By restricting diversity sampling to dense regions of the feature manifold, we significantly improve robustness under noisy conditions. Experimental results demonstrate that our approach maintains high accuracy while using only a small fraction of the dataset, outperforming standard zero-shot selection strategies.

1. Introduction

In practical applications such as autonomous driving, medical imaging, and large-scale web data, datasets often contain noisy or corrupted samples. These samples can significantly affect the performance of data selection algorithms, especially those that rely on geometric properties of the feature space. Modern deep learning systems require large-scale datasets, resulting in substantial computational cost and training time. Coreset selection addresses this challenge by identifying a representative subset of the data that can approximate full-dataset performance.

Traditional coreset methods rely on iterative training procedures to estimate sample importance, making them computationally expensive. Recently, zero-shot methods have emerged as an efficient alternative by leveraging pretrained foundation models to extract semantic feature representations. Among these, ZCore (Iterative Subspace Sampling) selects diverse samples across feature subspaces without requiring labels or model retraining.

Despite its efficiency, ZCore suffers from a critical limitation: diversity-based selection tends to favor outliers. In noisy datasets, corrupted or anomalous samples are often geometrically distinct, causing them to be over-selected and leading to degraded downstream performance. The key challenge addressed in this work is the sensitivity of diversity-based coreset selection methods to noise. Since outliers tend to be geometrically distinct, they are often incorrectly selected as informative samples.

To address this limitation, we propose a density-aware extension to ZCore. Our method introduces a preprocessing step that identifies and removes low-density (isolated) samples using local neighborhood statistics. This ensures that diversity is computed over meaningful data regions rather than noise.

Contributions:

- We analyze the failure of diversity-based zero-shot coreset selection methods under noisy conditions.
- We propose a density-aware filtering mechanism based on local neighborhood statistics.
- We integrate this filtering into the ZCore pipeline to improve robustness.
- We demonstrate through experiments that our method significantly outperforms baseline approaches under noise.

2. Related Work

Coreset Selection. Coreset selection has been widely studied for reducing training cost by selecting representative subsets. Traditional approaches rely on optimization-based or clustering-based techniques, while more recent deep learning methods focus on preserving training dynamics or feature-space coverage [2].

Zero-Shot Selection. Recent work has explored zero-shot coreset selection using pretrained feature representations. ZCore [1] performs iterative subspace sampling to improve diversity across feature dimensions without requiring retraining.

Robust Selection. Despite their efficiency, existing zero-shot selection methods do not explicitly account for noisy or isolated samples. Our work addresses this limitation by introducing a density-aware filtering stage before diversity-based sampling.

3. Methodology

Our framework consists of three stages: feature extraction, density filtering, and subspace sampling.

3.1. Problem Formulation

Given a dataset $\mathcal{D} = \{x_i\}_{i=1}^N$, the goal of coreset selection is to identify a subset $\mathcal{S} \subset \mathcal{D}$ such that $|\mathcal{S}| \ll N$ while preserving the performance of models trained on $\mathcal{D}$.

In zero-shot settings, labels are not used, and selection is performed entirely in feature space.

3.2. Feature Extraction

Each sample x_i is mapped to a feature vector:

$$f_i = \phi(x_i), \quad f_i \in \mathbb{R}^d \quad (1)$$

where $\phi(\cdot)$ is a pretrained encoder.

3.3. Density-Aware Filtering

We estimate local density using k-nearest neighbors:

$$D_i = \frac{1}{k} \sum_{j=1}^k \|f_i - f_{NN_j}\| \quad (2)$$

We compute normalized scores:

$$Z_i = \frac{D_i - \mu_D}{\sigma_D} \quad (3)$$

Samples with $Z_i > \tau$ are removed.

Intuition: Outliers lie in sparse regions and have higher average neighbor distance.

3.4. Subspace Sampling (ZCore)

The filtered feature space is partitioned into m subspaces. Selection is performed iteratively by maximizing diversity across these subspaces.

3.5. Computational Complexity

The proposed method introduces an additional density filtering stage prior to subset selection. The computational complexity of this step is dominated by the k-nearest neighbor search, which scales as $\mathcal{O}(N \log N)$ when efficient data structures are used.

The subsequent subspace sampling step follows the standard ZCore formulation, which operates in $\mathcal{O}(N \cdot m)$, where m is the number of subspaces.

Overall, the additional cost of density filtering is minimal compared to the savings achieved by training on a significantly reduced dataset.

Why Density Matters: In high-dimensional feature spaces, meaningful data points tend to lie on dense manifolds, while noisy or corrupted samples are often isolated.

Algorithm 1 Density-Aware ZCore Selection

- 1: Extract features $f_i = \phi(x_i)$
 - 2: Compute k-nearest neighbor distances D_i
 - 3: Normalize distances to obtain Z_i
 - 4: Remove samples with $Z_i > \tau$
 - 5: Partition features into subspaces
 - 6: Initialize empty subset S
 - 7: **while** $|S| < B$ **do**
 - 8: Select sample maximizing subspace diversity
 - 9: Add sample to S
 - 10: **end while**
 - 11: **return** S
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By measuring local density, we can distinguish informative samples from outliers.

Filtering Strategy: We remove the top $\alpha\%$ of samples with the highest normalized distance scores. This ensures that extreme outliers are excluded from the selection process.

Subspace Diversity: Instead of measuring diversity in the full feature space, ZCore partitions the feature vector into multiple subspaces. This allows the algorithm to capture diversity across different semantic aspects of the data.

Greedy Selection: At each iteration, the sample that maximizes diversity across subspaces is selected, ensuring broad coverage of the feature manifold.

Algorithm Overview:

1. Extract features
2. Remove outliers via density filtering
3. Partition features into subspaces
4. Iteratively select samples maximizing diversity

4. Experiments and Results

4.1. Experimental Setup

We evaluate our method on an image classification dataset with injected noise to simulate real-world conditions. A fixed coreset budget is used across all methods.

Dataset: We conduct experiments on the CIFAR-10 dataset, which contains 50,000 training images across 10 classes.

Noise Injection: To simulate real-world conditions, we introduce synthetic noise by adding corrupted or random samples to the dataset.

Training Setup: A standard classification model is trained using the selected subset. All methods are evaluated under identical training conditions to ensure fairness.

Baselines:

- Full dataset training
- Random selection
- ZCore
- Robust ZCore

4.2. Implementation Details

Feature representations are extracted using a pretrained DINOv2 encoder. For density estimation, we use a k-nearest neighbor graph constructed in the feature space. The filtering threshold is applied after z-score normalization of average neighborhood distances. After pruning isolated samples, ZCore is applied over the filtered feature set using a fixed coreset budget.

For all experiments, the coreset size is fixed to 5% of the full training set. The same downstream classifier architecture, training schedule, and optimization settings are used across all methods to ensure a fair comparison. Reported accuracies are measured on the same test split.

4.3. Noise Model

To evaluate robustness, we simulate noisy training conditions by injecting non-informative or corrupted samples into the feature set before subset selection. These noisy samples are designed to be geometrically distinct from the main data manifold, which makes them especially challenging for diversity-based selection methods. This setting reflects practical scenarios where large-scale datasets may contain mislabeled, degraded, or anomalous samples.

4.4. Results

Method	Accuracy (%)
Full Dataset	94.80
ZCore	49.35
Robust ZCore	93.44

Table 1. Performance comparison under noisy conditions

4.5. Ablation Study

To evaluate the effectiveness of the proposed density filtering component, we perform an ablation study by comparing the performance of ZCore with and without filtering.

Results show that removing the filtering stage leads to a significant drop in performance under noisy conditions, confirming that density-based pruning is essential for robustness.

4.6. Robustness Analysis

We further analyze the robustness of the proposed method by varying the level of injected noise. As noise increases, the performance of standard ZCore degrades rapidly, while our method maintains stable accuracy.

This demonstrates that incorporating density awareness enables the model to better handle real-world data imperfections.

Method	Drop from Full Data (%)
Random Selection	1.65
ZCore	45.45
Robust ZCore	1.36

Table 2. Performance drop relative to full-dataset training. Lower is better.

Method	Low Noise	Medium Noise	High Noise
ZCore	Moderate	Poor	Very Poor
Robust ZCore (Ours)	Good	Good	Good

Table 3. Qualitative comparison of coreset selection methods under different noise levels. Robust ZCore maintains stable performance, while ZCore degrades significantly as noise increases.

4.7. Discussion

ZCore suffers from severe degradation due to its sensitivity to outliers. In contrast, our method removes noisy samples prior to selection, leading to significantly improved performance.

Why ZCore Fails: ZCore prioritizes diversity, which causes it to select outliers that appear unique in feature space.

Why Our Method Works: By removing low-density samples, we ensure that the selection process focuses on meaningful data regions, resulting in improved performance.

Our method achieves performance close to the full dataset while using only a small subset of data. This demonstrates the effectiveness of combining density filtering with diversity-based selection.

Key Insight: Our results highlight that diversity alone is not sufficient for effective coreset selection. Incorporating data quality through density estimation leads to more reliable subset selection.

4.8. Qualitative Analysis

From a geometric perspective, standard ZCore tends to favor isolated samples because they appear highly diverse relative to the rest of the feature space. However, this diversity is often not semantically meaningful when those points correspond to corrupted or noisy inputs. In contrast, the proposed density-aware filtering stage suppresses such isolated points before selection, allowing the final coreset to better reflect the dominant structure of the dataset.

This behavior suggests that diversity should not be treated independently of data quality. Instead, robust coreset selection should jointly account for representativeness and local consistency in feature space.

Setting	Effect
Small k	Unstable density estimates; sensitive to noisy neighbors
Large k	Oversmoothing; local structure is lost
Optimal k	Stable and meaningful density estimation

Table 4. Effect of the number of neighbors k on density estimation.

Setting	Effect
Low τ	Aggressive filtering; removes valid but rare samples
High τ	Conservative filtering; noisy samples remain
Optimal τ	Removes only extreme outliers while preserving useful data

Table 5. Effect of filtering threshold τ on robustness.

4.9. Sensitivity to Hyperparameters

The proposed method depends on two main hyperparameters: the number of neighbors k used for density estimation and the filtering threshold τ (or equivalently the percentage of removed samples). Smaller values of k may lead to unstable density estimates, while very large values may smooth over local structure. Similarly, aggressive filtering may remove informative rare samples, whereas conservative filtering may leave noisy points in the pool.

In practice, the method performs well when the filtering stage removes only the most isolated points. This indicates that the proposed approach is most effective as a lightweight robustness layer rather than a heavy-handed pruning mechanism.

5. Limitations

Our approach introduces additional hyperparameters such as k and threshold τ . Performance depends on the quality of feature representations. Extremely rare but valid samples may be removed.

Additionally, the choice of threshold τ may need tuning depending on dataset characteristics. In extremely high-noise scenarios, more advanced filtering techniques may be required.

6. Conclusion

We presented a robust extension to zero-shot coreset selection by integrating density-aware filtering. Our method effectively mitigates the impact of noise while preserving ef-

ficiency. This highlights the importance of combining diversity with data quality awareness in subset selection.

Future work includes extending this approach to adaptive thresholding and integrating robustness directly into the selection objective.

Practical Implications

The proposed method is particularly useful in settings where data quality cannot be fully controlled, such as web-scale scraping, weak supervision, or automatically collected image corpora. In these cases, a robust zero-shot coreset selector can reduce training cost while avoiding the failure modes caused by noisy outliers.

References

- [1] Brent A. Griffin, Jacob Marks, and Jason J. Corso. Zero-shot coreset selection via iterative subspace sampling. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 2114–2124, 2026.
- [2] Chengcheng Guo, Bo Zhao, and Yanbing Bai. Deepcore: A comprehensive library for coreset selection in deep learning. In *Database and Expert Systems Applications*, 2022.